

FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform

6. Project Testing

Introduction

The testing phase validates the functionality, reliability, security, and performance of the FinRelief AI platform across all implemented modules.

Unit Testing

Individual modules including authentication, financial profile management, loan management, AI services, and history management were tested independently to verify expected functionality.

Integration Testing

Frontend and backend communication was verified using Axios and FastAPI APIs. Database connectivity and API responses were validated successfully.

System Testing

Complete end-to-end workflows including user registration, login, dashboard, loan management, financial health analysis, settlement prediction, negotiation email generation, borrower rights, and history modules were tested.

Error Handling Testing

The application was tested with invalid inputs, missing API keys, server exceptions, API failures, and unexpected responses. User-friendly error messages and fallback mechanisms were verified.

Performance Testing

Backend response times, JWT session handling, SQLite database performance, API communication, and concurrent request handling were evaluated for stable operation.

Test Results

All major functional modules operated successfully. Frontend and backend integration was verified, error handling worked as expected, and the platform demonstrated stable performance under different testing scenarios.

Conclusion

The FinRelief AI platform satisfied the planned functional requirements and demonstrated reliable, secure, and efficient operation suitable for academic project deployment.